Appendix

Flexible regulation as a double-edged sword: The case of the organic seed standard

Survey sampling method

Surveys were collected using a mixed of convenience and random sampling, with the approach varying across the three time periods. In the 2010 survey period, data were collected by convenience sample only, distributed by 23 organizations, including non-profits and foundations, organic certifiers, and departments of agriculture. In the 2015 survey period, 800 producers were randomly selected from the approximately 12,000 organic producers that use seed (e.g. not only fruit or livestock producers). From these 800, 203 responded, 127 of which were complete enough to use in our analyses. The remaining responses (672) were gathered through convenience sampling, where surveys were distributed by 41 organizations, again including non-profits and foundations, organic certifiers, and departments of agriculture. In the 202 survey period,, NEED TO SAY MORE.

Table 1: Convenience and random samples

Year	Total complete responses	Convenience responses (%)	Random responses (%)
2011	545	545 (100 %)	0 (0 %)
2016	799	672 (84 %)	127 (16 %)
2021	408	215 (53 %)	193 (47 %)

Survey representativeness

While the mix of sampling methods creates a challenge for determining the sample's representativeness of the population, we aim to demonstrate that our sample is representative by comparing the geographic distribution and approximate average farm size of our sample to USDA Agricultural Census data. We use USDA Census data from 2011, 2016, and 2021 to align with our 2010, 2015, and 2020 survey periods. We compare the number of operations in our sample (n) to the population (N) by region. PICK BACK UP HERE FOR 2021...

Across the three survey periods, the number of respondents represented 6.3%, 5.9%, and 2.5% of the total number of organic farmers identified in the USDA Census in the related years. For each survey period we calculated the standard deviation of the proportion of operations represented, which we used to identify regions that are over or under-represented (mean(% operations represented) +/- sd(% operations represented)). Over-represented year-region combinations are signaled by green text, while under-represented year-region combinations are signaled by red text. Only in 2011 do we see large variation in representation across geographies...

We also calculate the mean acreage in our sample, as well as the mean acreare according to the USDA census (N operations/total acreage) for a given year-region combination. Using the mean here provides limited information, given that farm sizes vary considerably. For instances, in our sample farm sizes in our sample are highly skewed, with many small farms and few really large farms. Similarly, county profiles from various states show ... (get country profiles with farm size bins). Though crude, these averages suggest relative similarity in average farm size.

Regarding concerns of sampling approach, we model the effect of sampling method with our 2020 survey data, including a term for responses that were gathered by random sampling compared to convenience sampling (Table 4). We see negligible changes int he results between the models run with and without the sampling consideration, and in the model in which sampling is considered the effect is small and not significant.

Table 2: Sample representativeness compared to USDA Censuses (2011, 2016, 2021)

Year	Region	Operations in sample (n)	Operations in USDA census (N)	Percent of operations represented (n/N)	Mean acreage in sample	Mean acreage in USDA Census	Mean acreage difference
2011	North Central	304	2986	10.2	195	212	-17
2011	Northeast	51	1897	2.7	142	142	0
2011	South	51	540	9.4	105	180	-75
2011	West	139	3296	4.2	194	248	-54
2016	North Central	395	4689	8.4	307	184	123
2016	Northeast	139	3246	4.3	142	135	7
2016	South	75	1096	6.8	144	215	-71
2016	West	190	4529	4.2	358	260	98
2021	North Central	177	5824	3	323	204	119
2021	Northeast	95	3888	2.4	205	143	62
2021	South	39	1209	3.2	847	279	568
2021	West	97	5273	1.8	329	284	45

Model selection and fit

Table 3: Comparison of model builds

	Proportion of organic land planted with conventional seed				
Forage crops (vs. Field)	0.819***			0.953***	0.919***
	(0.093)			(0.092)	(0.093)
Vegetable crops (vs. Field)	0.186**			0.524***	0.487***
	(0.076)			(0.102)	(0.103)
Western region (vs. North Central)	0.439***			0.321***	0.363***
	(0.085)			(0.084)	(0.085)
Northeastern region (vs. North Central)	0.325***			0.285***	0.285***
	(0.094)			(0.093)	(0.093)
Southern region (vs. North Central)	0.253**			0.151	0.161
	(0.115)			(0.113)	(0.113)
Year: 2015 (vs. 2010)	−0.371***			−0.220***	−0.207***
	(0.073)			(0.075)	(0.075)
Year: 2020 (vs. 2010)	−0.413***			−0.352***	−0.363***
	(0.086)			(0.090)	(0.092)
Certifiers request organic seed use					0.028
					(0.032)
Seed saved or traded (%)					−0.095***
					(0.031)
Values organic seed		−0.274***	−0.203***	−0.191***	−0.189***
		(0.033)	(0.033)	(0.033)	(0.033)
Farm size (log acres)		0.016	0.010	0.093***	0.096***
		(0.015)	(0.015)	(0.021)	(0.021)
Crop diversity		−0.054	−0.092*	−0.124**	−0.121**
		(0.053)	(0.052)	(0.054)	(0.054)
Barrier: Buyer requirements			0.092***	0.119***	0.116***
			(0.034)	(0.033)	(0.033)
Barrier: Insufficient quantity			0.025	0.001	0.0001
			(0.035)	(0.034)	(0.034)
Barrier: Undesirable traits			0.104***	0.109***	0.106***
			(0.037)	(0.036)	(0.036)
Barrier: Organic availability			0.236***	0.170***	0.170***
			(0.037)	(0.037)	(0.037)
Constant	−0.595***	−0.442***	−0.364***	−0.959***	−0.970***
	(0.068)	(0.087)	(0.088)	(0.145)	(0.144)
Observations	1,752	1,752	1,752	1,752	1,752
R^2	0.105	0.054	0.139	0.238	0.244
Log Likelihood	2,163.388	2,126.881	2,184.915	2,260.834	2,265.746

Note: *p<0.1; **p<0.05; ***p<0.01

Additional model considerations

Two sources of potential interpretation challenges include 1) the bias in the convenience sample and 2) the variation in certifiers.

Table 4: Considering sampling results for 2021 only

	Proportion of organic land planted with conventional seed	
Barrier: Organic availability	0.258***	0.257***
	(0.078)	(0.078)
Barrier: Buyer requirements	0.118*	0.119*
	(0.069)	(0.069)
Barrier: Undesirable traits	0.114	0.115
	(0.079)	(0.079)
Barrier: Insufficient quantity	0.025	0.025
	(0.079)	(0.079)
Values organic seed	−0.150*	−0.151**
	(0.077)	(0.077)
Farm size (log acres)	0.061	0.060
	(0.049)	(0.049)
Crop diversity	−0.125	−0.132*
	(0.079)	(0.074)
Seed saved or traded (%)	−0.098	−0.098
	(0.080)	(0.080)
Certifiers request organic seed use	−0.026	−0.025
	(0.068)	(0.068)
Forage crops (vs. Field)	0.498**	0.496**
	(0.204)	(0.204)
Vegetable crops (vs. Field)	0.001	−0.010
	(0.170)	(0.165)
Western region (vs. North Central)	0.356**	0.351**
	(0.179)	(0.178)
Northeastern region (vs. North Central)	0.312*	0.311*
	(0.170)	(0.170)
Southern region (vs. North Central)	0.083	0.080
	(0.239)	(0.239)
Random data collection (versus convenience)	0.040	
	(0.142)	
Constant	−0.898***	−0.875***
	(0.145)	(0.121)
Observations	408	408
R^2	0.216	0.216
Log Likelihood	606.308	606.269
Note:		*p<0.1; **p<0.05; ***p<0.01

Table 5: Considering certifiers

	Proportion of organic land planted with conventional seed	
Barrier: Organic availability	0.156**	0.159**
	(0.070)	(0.070)
Barrier: Buyer requirements	0.087	0.087
	(0.062)	(0.061)
Barrier: Undesirable traits	0.042	0.040
	(0.069)	(0.069)
Barrier: Insufficient quantity	0.024	0.023
	(0.064)	(0.064)
Year: 2015 (vs. 2010)	−0.298	−0.345**
	(0.240)	(0.138)
Year: 2020 (vs. 2010)	−0.394	−0.450*
	(0.383)	(0.255)
Values organic seed	−0.164***	−0.163***
	(0.063)	(0.063)
Farm size (log acres)	0.150***	0.153***
	(0.044)	(0.043)
Seed saved or traded (%)	−0.252**	−0.260**
	(0.106)	(0.104)
Certifiers request organic seed use	−0.083	−0.070
	(0.063)	(0.060)
Forage crops (vs. Field)	0.859***	0.843***
	(0.162)	(0.159)
Vegetable crops (vs. Field)	0.678***	0.651***
	(0.206)	(0.201)
Western region (vs. North Central)	0.397	0.362**
	(0.274)	(0.179)
Northeastern region (vs. North Central)	0.384	0.340
	(0.255)	(0.236)
Southern region (vs. North Central)	0.290	0.059
	(0.395)	(0.289)
Certifier 7	0.079	
	(0.385)	
Certifier 42	0.249	
	(0.400)	
Certifier 34	0.068	
	(0.334)	
Certifier 27	0.066	
	(0.267)	
Certifier 24	0.001	
	(0.398)	
Certifier 2	−0.278	
	(0.419)	
Constant	−1.117***	−1.008***
	(0.426)	(0.274)
Observations	487	487
R^2	0.240	0.238
Log Likelihood	625.536	624.823
Note:		*p<0.1; **p<0.05; ***p<0.01

Full model results

Table 6: Final model results

	Proportion of organic land planted with conventional seed
Barrier: Organic availability	0.170^{***}
	(0.037)
Barrier: Buyer requirements	0.116^{***}
	(0.033)
Barrier: Undesirable traits	0.106^{***}
	(0.036)
Barrier: Insufficient quantity	0.0001
	(0.034)
Year: 2015 (vs. 2010)	-0.207^{***}
	(0.075)
Year: 2020 (vs. 2010)	-0.363^{***}
	(0.092)
Values organic seed	-0.189^{***}
	(0.033)
Farm size (log acres)	0.096^{***}
	(0.021)
Crop diversity	-0.121^{**}
	(0.054)
Seed saved or traded (%)	-0.095^{***}
	(0.031)
Certifiers request organic seed use	0.028
	(0.032)
Forage crops (vs. Field)	0.919^{***}
	(0.093)
Vegetable crops (vs. Field)	0.487^{***}
	(0.103)
Western region (vs. North Central)	0.363^{***}
	(0.085)
Northeastern region (vs. North Central)	0.285^{***}
	(0.093)
Southern region (vs. North Central)	0.161
	(0.113)
Constant	-0.970^{***}
	(0.144)
Observations	1,752
R^2	0.244
Log Likelihood	2,265.746
Note:	*p<0.1; **p<0.05; ***p<0.01

Predicted conventional seed use

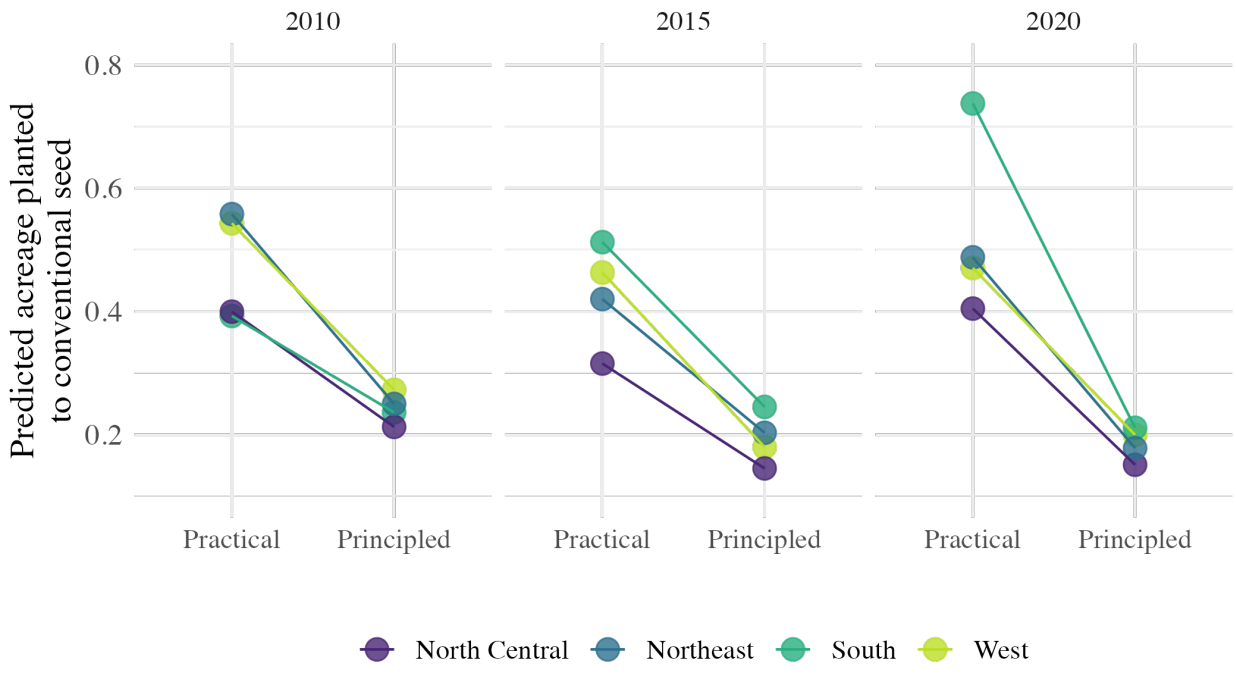


Figure 1: Predicted conventional seed use for field crop growers of two farmer profiles.

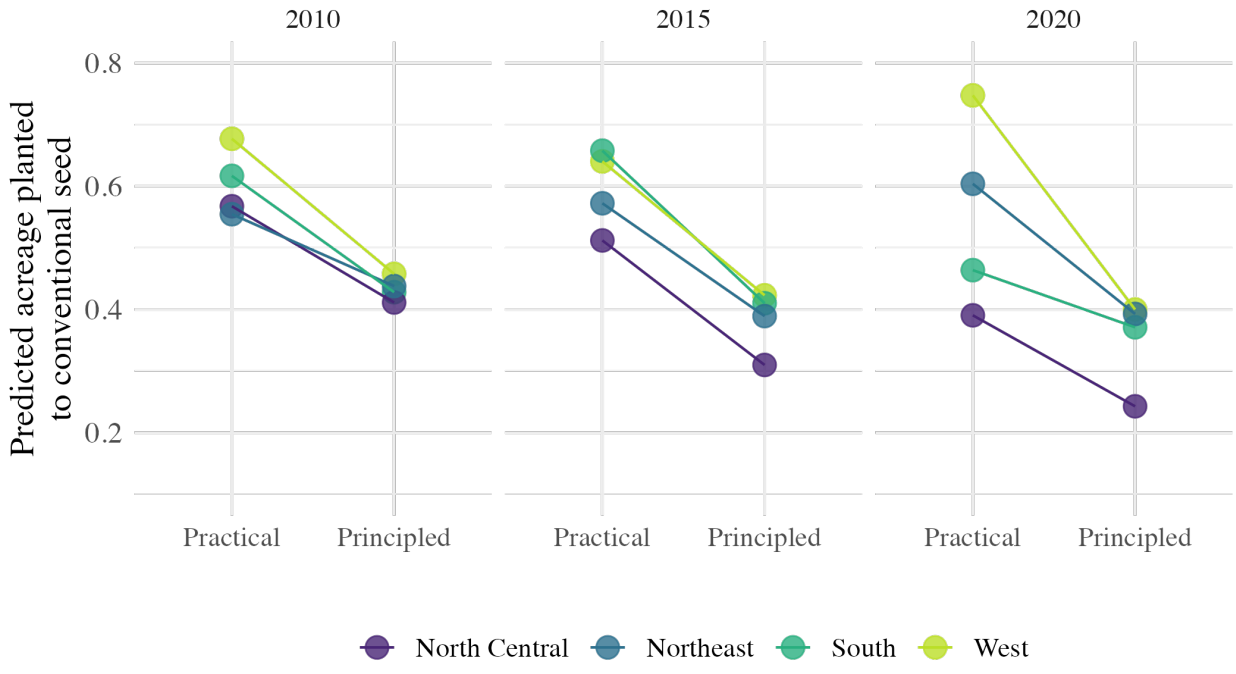


Figure 2: Predicted conventional seed use for forage crop growers of two farmer profiles.